

Heat Waves Are Frying Animal Brains -- and the Fallout Could Hit Our Food Supply

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On a scorching day in South Africa's Kalahari Desert, southern pied babblers face a simple test: a mealworm behind a transparent barrier. In cool weather, the birds quickly learn to go around. But when the mercury climbs, they just keep pecking at the plastic their problem-solving skills fried by the heat. This experiment is part of a growing body of evidence that extreme temperatures muddle animal minds. From birds to bees to fish, heat waves impair learning, memory, and decision-making. And the consequences could ripple far beyond the animal kingdom. Consider the humble bumblebee. At 77°F, most bees learn to associate a color with a sugary reward. At 90°F, fewer than half succeed. If pollinators forget which flowers to visit, crops like tomatoes and blueberries could suffer. "Changes in air temperature will affect brain temperature," says neuroscientist Emily Baird. A hotter brain may hinder nerve function, disrupting sensing, memory, and learning. Aggression also spikes with the thermostat. A 2023 study of nearly 70,000 dog bite reports across eight U.S. cities found incidents were 10% more likely on 90°F days than on 60°F days and not just because more people are outside. Researchers controlled for seasonal effects and still saw the pattern. "It's likely that both humans and dogs get stressed and more irate at higher temperatures," says neuroscientist Clas Linnman. Wild animals aren't immune. In the Italian Apennines, chamois goat-like mammals become **territorial** and **combative** as heat shrinks their food supply. Based on simulations, scientists predict chamois aggression could rise 50% by 2080 due to climate change. Even **vigilance** against predators falters. When Ridley's team tested pied babblers with a **taxidermied** genet (a cat-like predator), the birds reacted with alarm calls and flight but only in cooler weather. In extreme heat, they responded no differently to the predator than to a wooden box. "If the animals can't stay alert enough to find food or avoid predators, their chances of survival go downhill," says behavioral ecologist Amanda Ridley. For cold-blooded creatures like fish and insects, the effects may be especially dire. Male guppies in 90°F water struggle to navigate mazes, even with a mate as a reward. Mice overheat and forget objects they saw the day before. Heat waves can even accelerate biological aging in humans, speeding up "**epigenetic clocks**" that reflect stress on DNA. As climate change makes extreme heat more frequent, **cognitive impairments** across species could destabilize ecosystems. A sharp mind is increasingly vital for survival and the heat is making that harder for everyone.

Word Treasure

cognitive impairments -- Problems with thinking, learning, or remembering that can happen when the brain overheats.

epigenetic clocks -- Measurements that show how stress and environment can speed up biological aging by changing how DNA works, like a wear-and-tear timer.

vigilance -- The state of being watchful and alert to danger, which many animals lose in extreme heat.

territorial -- Behaving aggressively to defend a home area from other animals.

- combative** -- Ready or likely to fight, as hungry chamois become when heat shrinks their food supply.
- taxidermied** -- An animal that has been preserved by professional stuffing to look alive, used in experiments to test how real animals react to a predator.

Background you need

The link between heat and behavior isn't new - doctors have long noted that hospital visits for mental health and violent incidents rise during heatwaves. But recent studies zoom in on the brain's vulnerability. In the Kalahari Desert, southern pied babblers (small social birds) are part of a growing body of research led by behavioural ecologist Amanda Ridley. Her work shows that when body temperature spikes, even simple problem-solving can collapse. This matters far beyond birdwatching: if bees cannot learn which flowers to visit, crops like tomatoes and blueberries could produce less fruit. With climate models projecting more frequent and intense heatwaves, understanding these cognitive limits has become urgent for food security and conservation alike.

Neuroscientist Emily Baird explains that warmer air directly affects brain temperature, which can disrupt nerve messages. Meanwhile, a separate 2023 analysis of nearly 70,000 dog bite incidents across eight U.S. cities found a clear 10% jump on 90°F days, even after accounting for more time outdoors. Researchers like Clas Linnman think stress in both dogs and people is the culprit. In the Italian Apennines, chamois - goat-like mammals - already show increased territorial fights as heat shrinks their food supply, a trend that could worsen dramatically. The evidence is building that heat dims the spark of intelligence across the animal kingdom.

Break it down

LEAD: A vivid scene - pied babblers fail a simple problem-solving test in extreme heat

+/- specific: 'they just keep pecking at the plastic -- their problem-solving skills fried'

KEY CONCEPT: High temperatures impair animal cognition - learning, memory, decision-making

EVIDENCE 1 - POLLINATORS: Bumblebees cannot learn colour-reward associations at 90°F

+/- CONSEQUENCE: If bees forget which flowers to visit, crops like tomatoes and blueberries suffer

EVIDENCE 2 - AGGRESSION IN CITIES: Dog bite incidents 10% more likely on 90°F days across eight U.S. cities

+/- EXPLANATION: Stress in both dogs and humans rises with temperature

EVIDENCE 3 - WILD MAMMAL AGGRESSION: Chamois become territorial and combative as food shrinks

+/- PROJECTION: Aggression could rise 50% by 2080 due to climate change

EVIDENCE 4 - VIGILANCE FAILURE: Pied babblers fail to react to a taxidermied predator in heat

+/- SURVIVAL IMPACT: Inability to detect predators or find food reduces survival chances

BROADER TAXA: Cold-blooded animals (fish, insects) suffer even more - guppies struggle in maze, mice forget objects

HUMAN TOLL: Heat waves may speed up 'epigenetic clocks' that accelerate biological aging

CONCLUSION: Sharp minds are needed for survival, and climate-driven heat is making that harder for everyone

Why it matters

This isn't just about animals; when heat muddles pollinator brains, our food crops are at risk.

For a 12-year-old, the world they inherit could face more food shortages and more conflicts--both among humans and wildlife--if we don't address extreme heat's hidden damage to thinking itself.

Test what you remember

1. In the Kalahari experiment, what did pied babblers fail to do when it was very hot?
 - (A) Find the mealworm behind a transparent barrier by going around
 - (B) Recognize their own reflection
 - (C) Fly faster than usual
 - (D) Share food with younger birds
2. At what temperature did fewer than half of bumblebees learn to associate a colour with a reward?
 - (A) 50°F
 - (B) 77°F
 - (C) 90°F
 - (D) 100°F
3. According to a 2023 study, how much more likely were dog bites on a 90°F day compared to a 60°F day in U.S. cities?
 - (A) 5% more likely
 - (B) 10% more likely
 - (C) 20% more likely
 - (D) 50% more likely
4. What animal is predicted to become 50% more aggressive by 2080 due to climate change, based on simulations?
 - (A) Guppies
 - (B) Southern pied babblers
 - (C) Chamois
 - (D) Bumblebees
5. How did pied babblers react to a taxidermied predator in cool weather versus extreme heat?
 - (A) They always ignored it
 - (B) They gave alarm calls and fled in cool weather, but did not react differently from a wooden box in heat
 - (C) They attacked the predator in cool weather and ran away in heat
 - (D) They were more alert in hot weather
6. What effect can heat waves have on humans, according to the article?
 - (A) They only make people sweat more
 - (B) They can speed up biological aging measured by epigenetic clocks
 - (C) They improve memory temporarily
 - (D) They have no effect on human cognition

Answer key (try the quiz first!): 1: A 2: C 3: B 4: C 5: B 6: B

